

South Korea Financial Crisis (1997-1998).

A Macroeconomic and Time-Series Empirical Analysis.

Christopher Ajayi

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1.0 Introduction

The South Korean economy in mid-1997 experienced a painful blow after recording consistent economic growth rate averaging 8 percent for almost 30 years through high level of industrialization, export expansion and large-scale investment. Considered as one of the fastest growing economies in East Asia from the early 1960s to late 1990s, the South Korean economy was supported majorly by government backed conglomerates known as Chaebols, including Samsung, Daewoo, Hyundai and Kia. These conglomerates experienced a high level of expansion through unrestricted and unregulated borrowing of short-term loans obtained through domestic banks and other financial institutions.

Before the crisis, the Korean government partly or wholly owned specialized and development banks that accounts for about 17 percent of the country's financial system assets, and the role of these banks is to provide funds to specific and strategic economic sectors. Through the specialized and industrial banks, the government could provide funds for agricultural, fisheries and livestock cooperatives while the development banks supply long-term credit to major industries. However, these banks were subject to less scrutiny, supervision and global financial monitoring compared to other commercial banks regulated by the Korean Ministry of Finance and Economy ([Balino and Ubide, 1999](#)).

During the era of economic boom, there was also the case of Nonbank Financial Intermediaries (NBFIs) such as investment institutions, savings institutions and insurance companies. These NBFIs which constitute about 30 percent of the country's financial system asset towards the end of 1997, were directly or indirectly owned by the conglomerates and some other influential shareholders. The NBFIs acted as an intermediating chaebol and note on paper and were used to finance activities within the conglomerates including the use of trust accounts at commercial banks to dodge various restrictions on commercial bank lending ([Heo and Kim, 2000](#); [Balino and Ubide, 1999](#)). Prior to the financial crisis, there were about 30 merchant banks in operation, most of which were run by chaebols, and some were jointly run-in partnership with foreign investors to provide short term finance with less regulation, supervision and restrictions ([Balino and Ubide, 1999](#)).

South Korea's regulatory and supervisory flaws were revealed during the shift to a liberalized capital market and deregulated financial sector. Risky practices and regulatory arbitrage were promoted by the absence of centralized oversight, especially in merchant banks and trust accounts. These practices include poor accounting practices, weak loan classification standards, and regulators' ability to waive prudential rules ([Lee and Rhee, 2007](#)).

By 1997, the Korean economy has been dominated by heavy borrowing corporations with little or no profitability which in turn made them vulnerable to any disruption in cash flow. Also, the unregulated financial liberalization provided banks with freehand to incur excessive risk without sufficient financial strength to withstand shocks. [Hak Pyo \(2004\)](#) notes that the financial recklessness was exploited by the Chaebols to foster investment expansion in order to increase their market share in critical export market with little or no profits. Some conglomerates even diversified into new industries which were often unrelated to the business of the group ([Balino and Ubide, 1999](#)).

This report presents an empirical analysis of what happened before, during and after the crisis. The analysis presented in this report discusses the performance of macroeconomic variables in three sections. Section one evaluates the performance of these variables ten years before the financial crisis (1987-1997). Section two primarily focuses on the empirical findings associated with the near-term events prior to the crisis including background information about the crisis and measures taken to end the crisis (early 1997 – late 1998). In the final section, a re-examination and analysis of the macroeconomic variables directly linked to the crisis is carried out five years after the crisis (1999 – 2003). The data analysed in this report have been sourced from Federal Reserve Bank of St. Louis (FRED), International Monetary Fund (IMF) and World Bank. Some of the data collected are quarterly while some are annually. Description of data is presented in [Appendix 1](#).

2.0 Evolution of Key Macroeconomic Indicators Before the Korean Financial Crisis

2.1 Domestic Macroeconomic Indicator of Economy Heating Before the Crisis

This section shows the macroeconomic imbalances that made South Korea vulnerable before the 1997 Asian financial crisis.

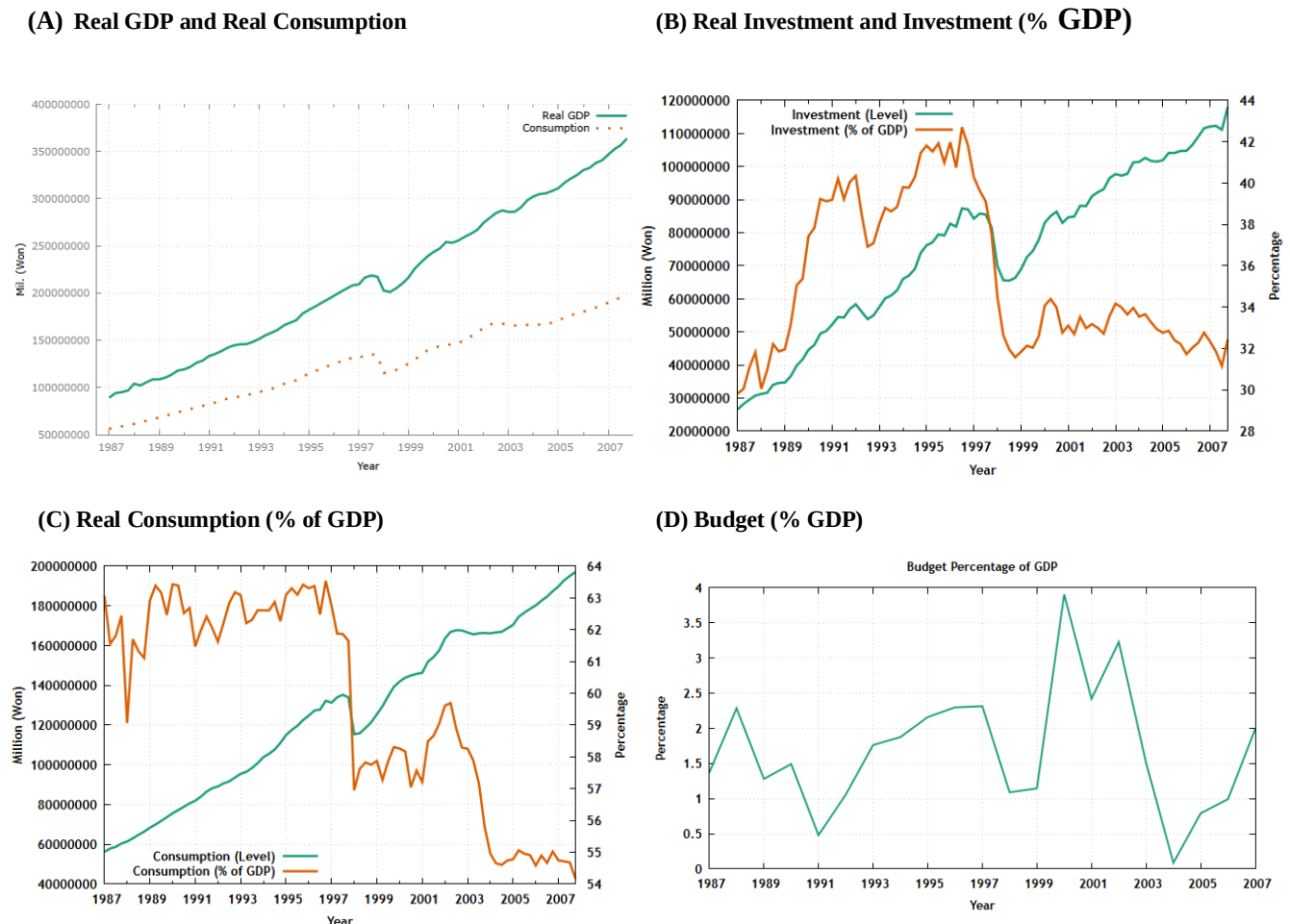


Figure 1: Macroeconomic Indicator of Domestic Economy Heating

Source: FRED, IMF and World Bank.

Figure 1 above shows the domestic macroeconomic condition of economic activities in South Korea before, during and after the crisis. However, our focus in this section is to discuss the domestic economic activities ten years before the crisis. **Panel A** shows that real GDP and private consumption grew at a steady state, nearly parallel upward trajectory, indicating that economic growth was tightly coupled with rising domestic living standards. From **Panel B**, real investment as a percentage of GDP (dotted line) rose aggressively from 1987, peaking at an exceptionally high 42 – 43 percent in 1996 - 1997. This highlights the heavily debt-leveraged corporate expansion (largely by South Korea's chaebols) that preceded the crisis. **Panel C** also

shows that from 1987 to 1997, private consumption was a highly stable component of the macroeconomy, consistently hovering between 61% and 63.5% of GDP. **Panel D** reveals the level of fiscal discipline throughout the late 1980s evidenced by a continuous surplus of about 2.5 percent in 1988, followed by a sharp decline to 0.5 percent in 1991. However, from 1991, budget surplus sharply increased to almost 2.5 percent in 1997 indicating fiscal strength. (Balino and Ubide, 1999).

2.2 Macroeconomic Indicator of External Financial Vulnerability

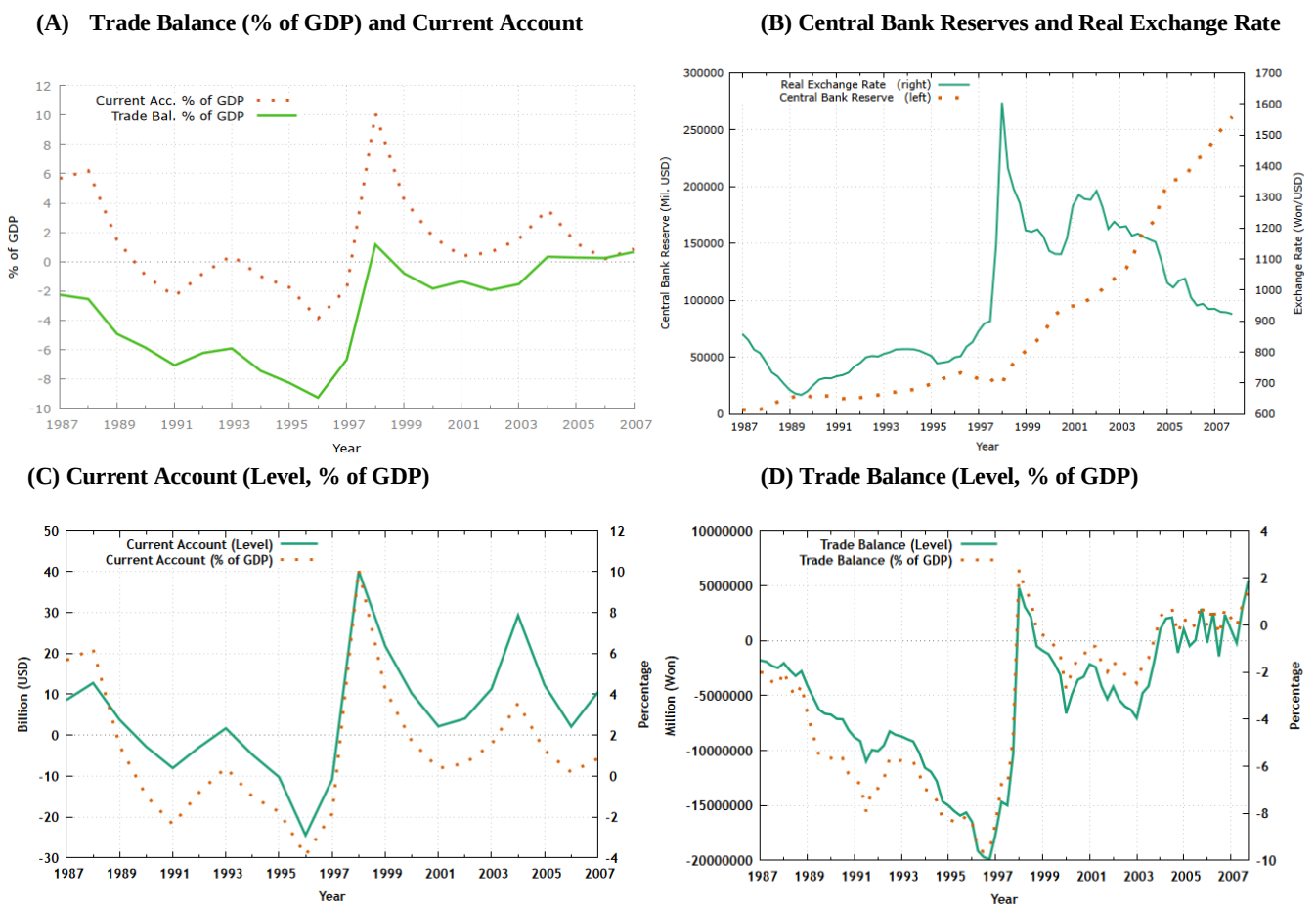


Figure 2: Macroeconomic Indicator of External Vulnerability

Source: FRED, IMF and World Bank.

Figure 2 above shows the level of external vulnerability of South Korean economy before the crisis. **Panel A** shows both trade balance and current account as percentage of GDP were negative from 1989-1996. Trade deficit increased drastically at almost nine percent of GDP in 1996, showing that South Korea was consuming and investing far more on foreign goods than it was producing as exports. **Panel B** shows the real exchange rate (Won/USD). The graph shows a sudden spike in the exchange rate from

around 800 Won/USD to over 1600 Won/USD in mid-1997. This represents a massive, rapid devaluation of the Korean won. Panel C shows that the current account steadily deteriorated into a deep deficit nearly about \$25 billion USD in 1996. Meaning that South Korea was a net borrower from the rest of the world, domestic investment was greater than savings, hence the need for increased borrowing of capital from the rest of the world, and consequently accumulating short-term foreign-currency debt. From Panel D, trade deficit in Won worsened progressively, reaching a massive deficit of about ₩20 trillion by late 1996. This indicates heavy reliance of the conglomerates on imported raw materials to promote domestic investment without a corresponding revenue from export to balance the account.

2.3 Macroeconomic Indicator of Financial Fragility Level

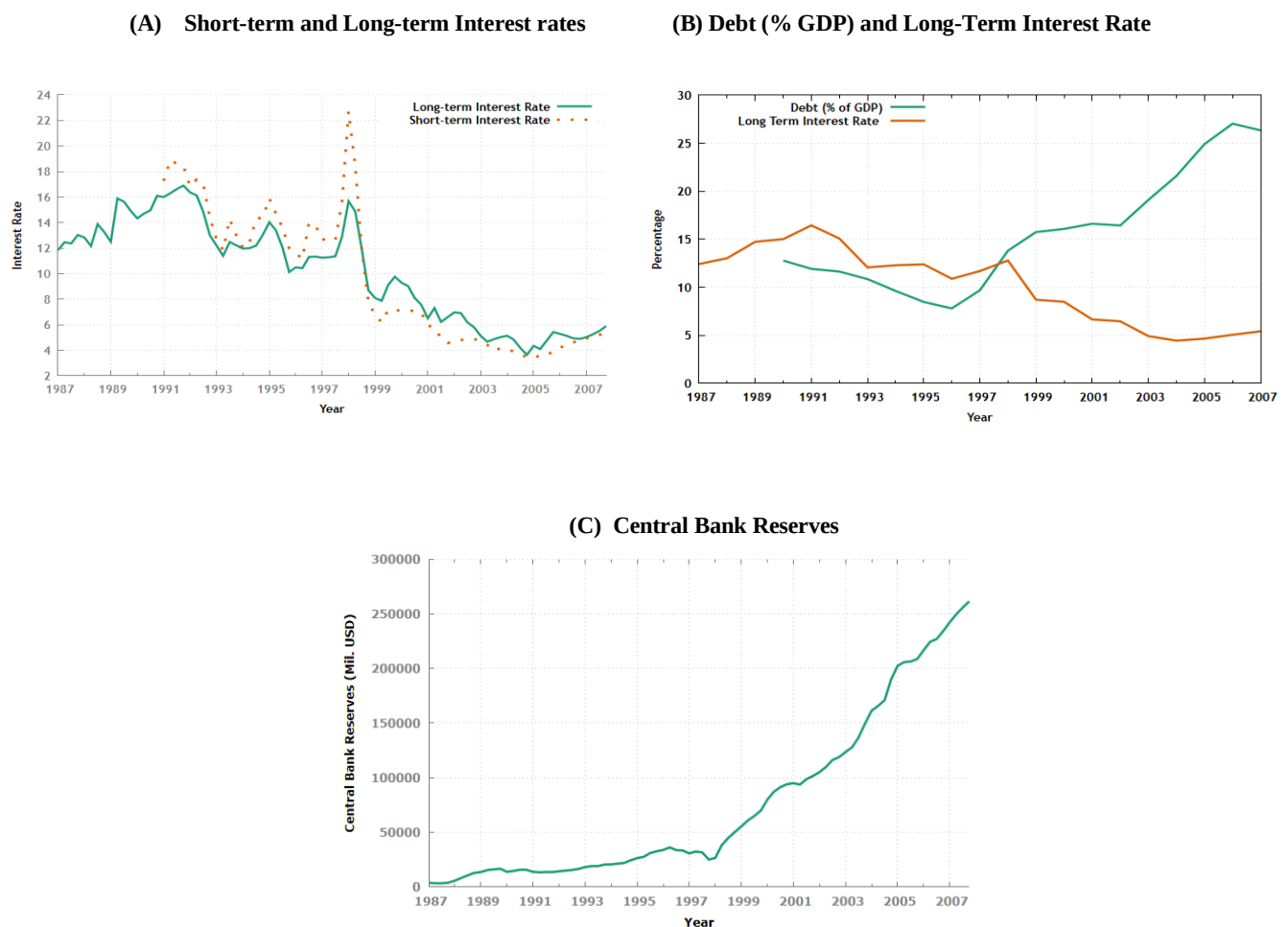


Figure 3: Macroeconomic Indicator of Financial Fragility

Source: FRED, IMF and World Bank.

Figure 3-Panel A shows that interest rates were persistently high, ranging between 10 – 17 percent. Short-term interest is seen to hover above or near long-term interest rates. South Korea maintained this level of high interest rate as a measure to curtail inflation during the rapid economic expansion in the period leading to the financial crisis. However, when the crises struck, short-term interest rate spiked, peaking over 22 percent. **Panel B** shows that during the early 90s, South Korea maintained a reduced national debt as a percentage of GDP from 13% in 1990 to about 8 percent by 1996. The state was highly fiscally responsible. **Panel C**, reveals that for a decade before the crisis, South Korea maintained a very low foreign exchange reserves below \$50 billion USD. Between late 1996 and late 1997, there was further reduction in the foreign exchange reserve as investors panicked and liquidated their investment and the Bank of Korea was trying to defend the Won against the dollar which led to the exhaustion of the small reserve that was available (Balino and Ubide, 1999).

3.0 Timeline of the Crisis

3.1 Near-Term Events Prior to South Korea Financial Crisis

South Korea's economy grew at an average rate of more than 8% in the early 1990s because of the combination of industrialization and globalization which led to an export economy. However, growth became increasingly unsustainable in early 1997 due to the aggressive expansion of major conglomerates like Daewoo, Kia Motors, and Hyundai, consequently leading to a surge in real investment and investment as a percent of GDP (see Figure 1: Panel B). The aggressive expansion of the conglomerates (popularly referred to as the Chaebols) was majorly financed by debt rather than retained profits (Kruger and Yoo, 2002). These Chaebols leveraged excessive short-term foreign borrowing which made the financial system more vulnerable. At comparatively low short-term interest rates, Korean commercial banks and merchant banks took out large loans from foreign lenders and used the money for domestic business lending (Balino and Ubide 1999). In order to sustain economic growth and financial stability, South Korea became more reliant on ongoing foreign capital inflows as its external debt grew during the mid-1990s.

Prior to the crisis, external imbalance indicators such as current account declined. Between 1993 and 1996, the current account deficit increased significantly due to robust import growth and growing need

for outside funding. Similarly, exchange rates and reserves provide evidence that the Korean won was under intense pressure in the early quarter of 1997 while foreign exchange reserves progressively decreased. Early in 1997, a number of significant Chaebol's bankruptcy including Hanbo Steel, Sammi Steel and Kia Motors led to even more financial turbulence. The collapse of these companies dealt a severe blow not only to the financial institution that had been lending to them and also to the struggling economy which further led to a shock to foreign investors. These bankruptcies revealed flaws such as inadequate lending procedures and lax financial oversight in the banking industry (Lee, Kyu-Sung, 2011). Also, the Thai baht's fall in July 1997 exacerbated East Asian financial contagion and due to their high external obligations and inadequate banking systems, the Asian economies' financial stability including South Korea was re-evaluated by foreign investors. Capital outflows from South Korea increased as confidence declined, putting more pressure on the Korean won to depreciate and limiting access to foreign credit markets (Corsetti et al., 1999).

Additionally, Kim Kihwan (2006) shows that internal weaknesses and external shocks contributed to the 1997–1998 Korean financial crisis. A significant contributing element was the rise of the U.S. dollar starting in 1995 against East Asian currencies including about 12 percent increase against the South Korea Won, which raised non-performing loans in the banking sector, expanded the trade deficit, hampered corporate profitability, and decreased Korea's export competitiveness against Japan (Radelet and Sachs, 1998). A lack of liquidity in the area was also exacerbated by the Japanese yen's fall, which decreased Japanese investment in Korea and forced Japanese banks to cut off lending (Koo and Kiser 2001). The government's refusal to save financially challenged chaebols like Hanbo and the uncertainty surrounding how Kia's problems were handled cast doubt on the continuity of economic policy, which further amplified the domestic economic issues (Yoon Cho, 1998).

Similarly, Keun See Lee (1998) revealed that the Korean financial crisis was largely caused by a number of factors, including excessive borrowing brought on by industrial overexpansion, capital flight after financial unrest in other Asian economies, rising labor costs, delayed government intervention, close political-business ties, a focus on asset growth rather than profitability, ongoing trade imbalances,

reliance on short-term external debt, reckless lending practices by deregulated financial institutions, and a lack of financial transparency.

As the Southeast Asian financial crisis developed, international creditors withdrew funding and refused to roll over short-term loans to Korean banks, which increased these worries. Investor confidence plummeted as foreign exchange reserves fell quickly, hastening capital flight and finally compelling Korea to turn to the IMF for support in November 1997 ([Balino and Ubide, 1999](#); [Yoon Cho, 1998](#)).

3.2 Events During the Crisis

Between October and December 1997, the South Korean financial crisis dramatically worsened. Foreign lenders refused to roll over short-term loans owing by Korean banks and enterprises as investor confidence declined. Due to the financial system's heavy reliance on short-term external financing, this led to significant liquidity shortages ([Radelet and Sachs 1998](#); [Balino and Ubide, 1999](#)). Exchange rate data shows a significant decline in the value of the Korean won relative to the US dollar in late 1997 ([Keun See Lee 1998](#)), which consequently led to increased debt estimated around \$150 billion. In an effort to pay its external debt obligations and stabilize the currency, the Bank of Korea saw a sharp fall in its foreign exchange reserves. The government's capacity to protect the currency rate was diminished by reserve depletion, which heightened fear in the financial markets ([Corsetti et al., 1999](#); [Balino and Ubide, 1999](#))

[Martin Feldstein \(1998\)](#) notes that during the crisis, interest rate indicators also showed deteriorating financial conditions. As authorities tightened monetary policies to curb capital outflows and boost confidence, short-term interest rates rose dramatically. Higher interest rates, however, made highly leveraged companies even more vulnerable and raised the cost of debt servicing across board ([Kang et al., 2001](#)).

In addition, [Hsieh and Lee \(2020\)](#) finds that South Korea had almost completely lost the trust of the global market. International banks decreased their lending exposure to Korean financial institutions, while credit

rating agencies downgraded Korean government debt. This in turn led to significant capital flight (Jae-Hyun Suh, 2019), a drop in industrial production (OECD, 1999), and rising unemployment resulting from massive layoffs were all experienced by the economy (Coe and Kim 2002; Koo and Kiser 2001; Keun See Lee 1998).

3.3 Measures to end the crisis

In response to the financial crisis, the International Monetary Fund received a formal request for emergency financial assistance from the South Korean government in December 1997. The IMF authorized a rescue package valued at about \$58 billion which is considered to be one of the biggest bailout schemes at the time. The intervention mandated that South Korea carry out significant structural change and macroeconomic stabilization. From the \$58 billion, an initial \$35 billion was granted by the IMF and other multilateral organizations and an additional \$23.4 billion supplementary support facility to be supplied by the G-7 nations (Kim Kihwan 2006).

Subsequently, South Korea enacted extensive measures which includes the creation of a single system of financial supervision, improvement of the legal and regulatory environment by passing a number of financial reform laws that created the Financial Supervisory Commission (FSC) and the Financial Supervisory Service (FSS). In addition to this, the Korea Asset Management Corporation (KAMCO) was expanded to help resolve non-performing loans, and regulatory agencies were given more authority to reorganize problematic institutions (Kyu-Sung Lee, 2011).

Bank recapitalization, liquidity provision and removal of trouble assets of the balance sheets of financial institutions were also carried out by the government through the infusion of significant government spending into the banking sector. This led to the closure, merger and restructuring of numerous weak commercial banks, merchant banks, and other non-bank financial institutions thereby restoring confidence in the financial system (Kulam 2021). Following this, stricter prudential regulations were enacted by policymakers, with practical steps to lessen moral hazard. Additionally, stricter limits on short-term foreign borrowing, significant loan exposures, and disclosure requirements were put in place,

including updated asset categorization criteria to properly account for future credit risks (Balino and Ubide 1999).

Furthermore, in order to dispel the notion of unrestricted governmental guarantees and promote more market discipline among depositors, the government also implemented a partial deposit insurance program. Also, significant corporate governance changes and financial liberalization were also sought by South Korea (OECD, 1999) which led to the demise and reorganization of a number of sizable chaebols indicating the end of implicit government bailouts, which lowered expectations of state backing among lenders and borrowers and encouraged more responsible financial practices (Kulam, 2021). The country's integration into international financial markets was enhanced by the implementation of a floating exchange rate regime, the lifting of prohibitions on foreign investment, and the deregulation of the stock, bond, and real estate markets (Park et al, 2001). Simultaneously, governance reforms boosted managerial responsibility, expanded foreign involvement in financial institutions, and strengthened shareholder rights (Koo and Kiser 2001).

Together, these changes increased market liquidity, strengthened the financial system's resilience, increased the capitalization and profitability of financial institutions, and aided in the creation of more efficient risk-monitoring frameworks, such as stress testing and early warning systems intended to identify new financial vulnerabilities.

4.0 The Post Crisis Economic Trajectory

The government's intervention during the crisis was majorly focused on the factors associated with the crisis, such as fluctuating exchange rates, current account deficits, diminishing foreign reserves, excessive corporate debt, inadequate banking systems, and financial market volatility. Macroeconomic data suggests that many of these changes were mostly successful in re-establishing investor confidence and economic stability five years after the crisis. Prior to the crisis, South Korea was heavily dependent on short-term foreign capital inflows, evident by its current account deficits and diminishing foreign exchange reserves between January 1997 and late 1998.

Five years after the Asian Financial Crisis of 1997, South Korea appears to have recovered rather quickly, according to post-crisis macroeconomic data. The government's response to the financial crisis through monetary and exchange rate policies, fiscal policies, financial sector reforms, corporate sector restructuring, and structural and institutional reforms yielded some positive results five years down the line (Lee and Rhee, 2000). Data shows significant gains in economic growth, reserve accumulation, currency rate management, and external stability after IMF-backed stabilization and structural reform initiatives were put in place. Figure 4, 5 and 6 present a post-crisis re-evaluation of the macroeconomic variables 5 years after the financial crisis.

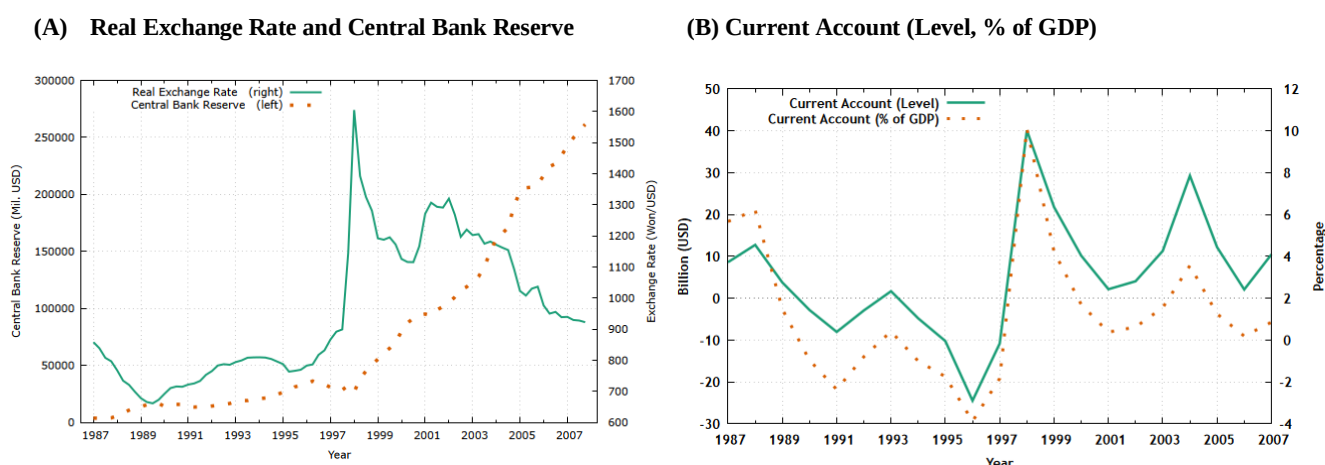


Figure 4: Real Exchange Rate, Central Bank Reserve and Current Account (Level, % of GDP)

Source: FRED, IMF and World Bank.

Following the crisis as shown in **Figure 4 (Panel A)**, South Korea steadily rebuilt trust in its foreign sector, according to indications of the central bank reserve and exchange rate. The Korean won fell drastically during 1997 and 1998, peaking at the height of the crisis as about 1600 Won, while moves by the government to protect the currency caused foreign exchange reserves to drastically decrease. Reserves did, however, rise steadily after 1998 during the time of recovery, suggesting improved reserve management and fresh capital inflows. Simultaneously, the rapid devaluation period was followed by a gradual stabilization of the currency. This implies that stricter monetary policy, IMF support, and exchange rate liberalization were mostly effective in boosting external confidence and lowering vulnerability to speculative attacks (Jounghyeon Kim 2012; Coe and Kim 2002).

Additionally, the current account graph in **Figure 4 (Panel B)** shows that South Korea's external situation significantly improved following the crisis. The nation's current account deficits prior to 1997 were a reflection of its growing reliance on foreign borrowing and capital inflows. However, the current account quickly moved into surplus following the crisis, reaching a peak of almost 10 percent of GDP in 1998 and above 4 percent in 2003. This reverse implies that while recessionary conditions decreased import demand, exchange rate depreciation increased export competitiveness. The balance was positive for the most of the post-crisis period, despite the current account surplus reducing in 2001, suggesting that external stability has continued to improve and that there is less need for short-term foreign funding.

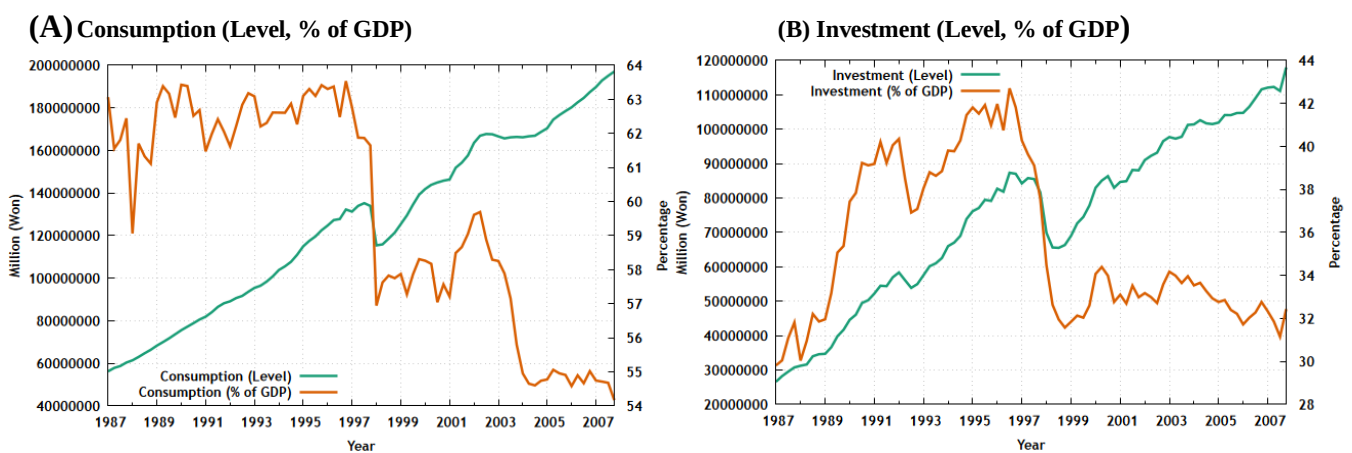


Figure 5: Consumption (Level, % of GDP) and Investment (Level, % of GDP).

Source: FRED, IMF and World Bank.

The consumption graph in Figure 5 (Panel A) above, shows that actual consumption levels fell steeply during the crisis, which was caused by deteriorating economic condition, uncertainty, increased unemployment, and declining domestic demand. But after 1999, consumption levels started to steadily rise again, a sign of a slow recovery in household spending and growing economic confidence. However, consumption as a percentage of GDP remained between 57 and 60 percent 5 years after the crisis, a level lower than the initial 64 percent before the crisis in early 1997, despite the recovery in absolute consumption. This implies that, in contrast to the pre-crisis era, post-crisis economic growth was less driven by consumption.

The contractionary stabilization policies put in place under the IMF and the structural adjustment program by the South Korean government are mostly responsible for the decline in consumption as a percentage of GDP. During the adjustment period, domestic demand was lowered by labour market restructuring, higher interest rates, and stricter credit requirements (Lee and Ree, 2007). There is also the possibility that consumer's increased savings after the crisis led to lower consumption in comparison to output growth.

Significant post-crisis structural adjustment is also shown by the investment metrics. Between 1997 and 1998, real investment levels drastically decreased as private-sector activity was hampered by financial instability, company bankruptcies, and tighter monetary conditions. Nonetheless, investment steadily increased after 1999 and persisted in doing so into the early 2000s, indicating improved macroeconomic stability and revitalized company confidence. This recovery implies that better financial supervision, company restructuring, and banking reforms put in place by the government helped to revive investment activity following the crisis.

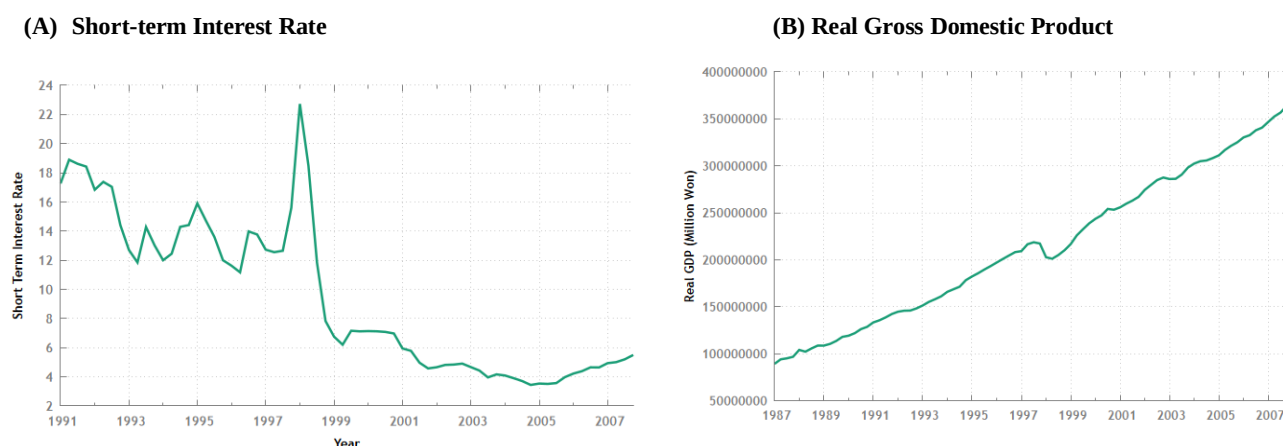


Figure 6: Short-term interest rate and Real GDP

Source: FRED, IMF and World Bank.

Figure 6 shows the result of the strong monetary tightening policies implemented throughout the crisis is reflected in the short-term interest rate graph. In an effort to stabilize the Korean won, curb capital outflows, and boost investor confidence, interest rates rose dramatically between 1997 and 1998 as shown in [Panel A](#). Short-term interest rates steadily decreased once financial markets stabilized and stayed

mostly steady during the recovery phase (Coe and Kim 2002). This suggests that investor confidence steadily increased and that monetary circumstances were brought under control following the crisis. However, the sharp rise in interest rates during the crisis probably made highly leveraged companies more financially distressed and added to the 1998 short-term recession (Furman & Stiglitz, 1998).

The real GDP graph in Panel B demonstrates how the crisis caused a brief economic recession in 1998 as a result of decreased domestic demand, financial instability, and corporate failures. But in 1999, GDP growth picked up again and continued to rise throughout the early 2000s. The comparatively quick rebound in output indicates that financial stability measures, export growth, and structural changes all played a major role in reviving economic activity. South Korea's economy recovered from the crisis at one of the quickest rates when compared to many other Asian economies (Lee and Rhee, 2000).

In conclusion, while in the early months of recovery, the adjustment process resulted in substantial short-term economic losses, especially due to high interest rates, recessionary pressures, and higher unemployment. Overall, the stabilization and structural reform initiatives implemented during the crisis appear to have been generally successful in addressing the significant imbalances that led to the crisis, according to the macroeconomic indicators following the crisis (Hak Pyo, 2004). Together, the current account balance improvement, foreign exchange reserve accumulation, exchange rate stabilization, consumption levels, GDP growth recovery, and interest rate normalization show that South Korea was able to successfully restore macroeconomic stability within five years of the crisis.

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Appendix 1

Quarterly (1987 – 2007)

- Real Gross Domestic Product
- Real Consumption (level, and % GDP)
- Real Investment (level, and % GDP)
- Real Trade Balance (level, and %GDP)
- Current Account (level, % of GDP)
- Exchange Rate
- Long-term interest rate
- Short-term interest rate (from 1991)

Yearly (1987 – 2007)

- Debt (% of GDP)
- Budget (% of GDP)
- Central Bank Reserves

Macroeconomic Indicator of External Vulnerability

Source: FRED, IMF and World Bank.